OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS

GRADE 7 MATHEMATICS

OKLAHOMA Education

1 **Which set of numbers is arranged in order from <u>least</u> to <u>greatest</u>?**

A $0.25, \frac{1}{3}, \frac{3}{5}, 0.85$

B $\frac{1}{3}, 0.25, \frac{3}{5}, 0.85$

C $0.25, 0.85, \frac{1}{3}, \frac{3}{5}$

D $\frac{1}{3}, \frac{3}{5}, 0.25, 0.85$

Standard: 7.N.1.1 Compare and order rational numbers expressed in various forms using the symbols <, >, and =.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to order numbers, which include both fractions and decimals, from smallest to largest.

Distractor Rationale:

A. Correct. The student demonstrated an ability to compare and order rational numbers expressed in various forms.

B. The student confused $\frac{1}{3}$ with 0.13.

C. The student did not know how to compare fractions to decimals so lists the decimals first and then the fractions.

D. The student did not know how to compare fractions to decimals so lists the fractions first and then the decimals.

2

Match each number in the left column with the equivalent number in the right column.
To connect numbers, click a number in the left column and then a number in the right column, and a line will automatically be drawn between them. To remove a connection, hold the pointer over the line until it turns red, and then click it. Each number in the left column matches to only one number in the right column.

$\frac{3}{4}$	$\frac{1}{4}$
$3.\overline{33}$	$\frac{3}{8}$
0.25	$\frac{1}{5}$
$\frac{2}{10}$	$\frac{9}{12}$
0.375	$\frac{10}{3}$

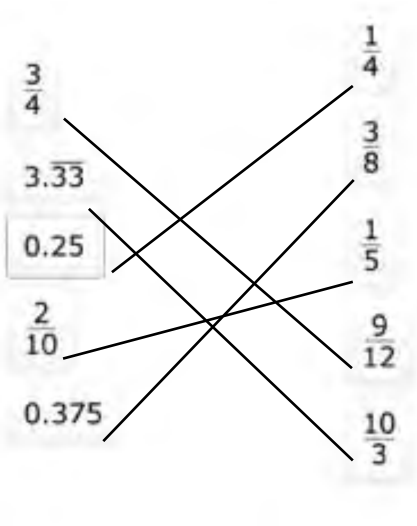
Standard: 7.N.1.2 Recognize and generate equivalent representations of rational numbers, including equivalent fractions.

Depth-of-Knowledge: 2

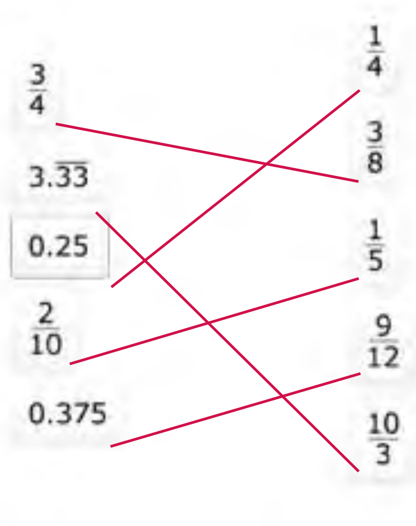
This item is a DOK 2 because it requires the student to recognize equivalent numbers in different forms, including fractions and decimals.

Sample Distractor Rationales:

Correct



Incorrect



The student thought fractions with the same numerator are equivalent.

3 **Gabrielle had $300 in her checking account and $125 in her savings account. If Gabrielle transferred enough money from her checking account to her savings account to double the savings account balance, what is her new checking account balance?**

A $350

B $250

C $175

D $125

Standard: 7.N.2.5 Model and solve problems using rational numbers involving addition, subtraction, multiplication, division, and positive integer exponents.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine how to approach the problem, and then solve a non-routine problem involving more than one operation.

Distractor Rationale:

A. The student found the amount that needed to be transferred, $125, then doubled it to get $250, and then added that to the existing $125.

B. The student doubled the savings account balance.

C. Correct. The student demonstrated an ability to solve a real-world problem involving addition, subtraction, and multiplication.

D. The student found the amount that needed to be transferred.

4 **An expression is shown.**

$$8 - 4 \bullet 3^3 + 11$$

What is the value of this expression?

A −89

B −17

C 23

D 119

Standard: 7.N.2.5 Model and solve problems using rational numbers involving addition, subtraction, multiplication, division, and positive integer exponents.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the value of an expression.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to solve a mathematical problem involving calculations with rational numbers and positive integer exponents.**

B. The student thought $3^3 = 3 \times 3 = 9$.

C. The student did not know how to solve for exponents or how to use order of operations and computed $(8 - 4) \times 3 + 11$.

D. The student computed $(8 - 4) \times 3^3 + 11$.

5

$$\frac{12+18}{3}+5-3^4$$

What is the value of this expression?

A -66

B -54

C 26

D 96

Standard: 7.N.2.5 Model and solve problems using rational numbers involving addition, subtraction, multiplication, division, and positive integer exponents.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the value of an expression.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to solve a mathematical problem involving calculations with rational numbers and positive integer exponents.**

B. The student computed $\frac{12}{3} + 18$ instead of $\frac{12+18}{3}$.

C. Balance distractor

D. The student added 3^4 instead of subtracted.

6 **Which equation represents a proportional relationship?**

A $y = -2x$

B $y = 5 - 2x$

C $y = 2x - 5$

D $y = 2x + 5$

Standard: 7.A.1.1 Identify a relationship between two varying quantities, x and y, as proportional if it can be expressed in the form $y/x = k$ or $y = kx$; distinguish proportional relationships from non-proportional relationships.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall a term or definition, proportional relationship.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to identify an equation that represents a proportional relationship.**
B. The student did not know how to determine if an equation represents a proportional relationship.
C. The student did not know how to determine if an equation represents a proportional relationship.
D. The student did not know how to determine if an equation represents a proportional relationship.

7 **Which table shows an inversely proportional relationship between *x* and *y*?**

A

x	*y*
5	60
10	30
15	20
20	15

B

x	*y*
5	3.5
10	7.0
15	10.5
20	17.5

C

x	*y*
5	−4
10	−11
15	−18
20	−25

D

x	*y*
5	15
10	30
15	45
20	60

Standard: 7.A.1.1 Identify a relationship between two varying quantities, *x* and *y*, as proportional if it can be expressed in the form $y/x = k$ or $y = kx$; distinguish proportional relationships from non-proportional relationships.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract information from the table in order to find the one that shows an inversely proportional relationship.

Distractor Rationale:

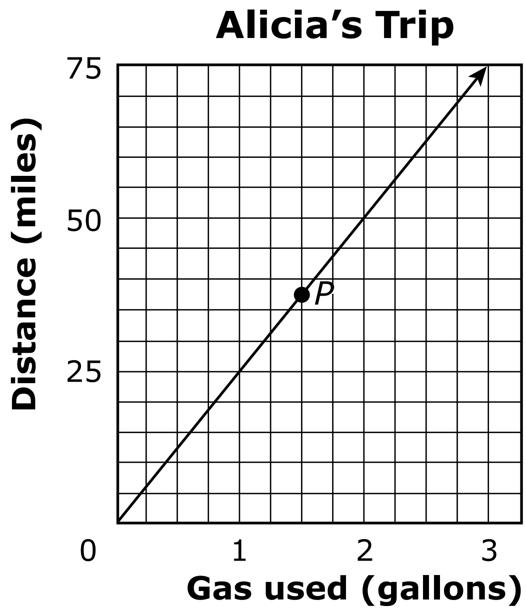
A. **Correct. The student demonstrated an ability to identify an inversely proportional relationship between *x* and *y*.**

B. The student confused proportional and inversely proportional relationships.

C. The student selected a relationship where the *y*-values decrease at a constant rate.

D. The student confused proportional and inversely proportional relationships.

8 **Alicia drove to her grandparents' house. The graph below shows the number of gallons of gas used and the distance traveled during the trip.**



According to the graph, which statement <u>best</u> describes point *P*?

A Alicia used 1.5 gallons to travel a distance of 37.5 miles.

B Alicia used 37.5 gallons to travel a distance of 1.5 miles.

C Alicia traveled at a rate of 37.5 miles per hour.

D Alicia traveled at a rate of 1.5 miles per hour.

Standard: 7.A.2.1 Represent proportional relationships with tables, verbal descriptions, symbols, and graphs; translate from one representation to another. Determine and compare the unit rate (constant of proportionality, slope, or rate of change) given any of these representations.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract information from the graph in order to find the statement that best describes a point on the graph.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to represent a proportional relationship with a verbal description.**

B. The student confused the x and y axes.

C. The student focused on the y-value and chose a common relationship.

D. The student focused on the x-value and chose a common relationship.

9 **Manuel is cooking dinner for his family of 4. The recipe he is using makes dinner for 6 people. The recipe calls for 3 cups of flour. How many cups of flour should Manuel use to make the recipe for 4 people?**

A 1 cup

B 2 cups

C 7 cups

D 8 cups

Standard: 7.A.2.2 Solve multi-step problems with proportional relationships (e.g., distance-time, percent increase or decrease, discounts, tips, unit pricing, mixtures and concentrations, similar figures, other mathematical situations).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to solve a multi-step problem.

Distractor Rationale:

A. The student computed $4 - 3$.

B. Correct. The student demonstrated an ability to solve a problem using proportional relationships.

C. The student computed $4 + 3$.

D. The student set up an incorrect proportion, $\frac{3}{6} = \frac{4}{x}$.

10 **Carla bought 8 equally priced theater tickets. The total cost was $190 including a $6 service charge. This equation can be used to find t, the price of each ticket.**

$$8t + 6 = 190$$

What is the price of each ticket, t?

A $14.00

B $23.00

C $23.75

D $24.50

Standard: 7.A.3.1 Write and solve problems leading to linear equations with one variable in the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are rational numbers.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, solving for a variable in a given equation.

Distractor Rationale:

A. The student computed 8 + 6 = 14 or $8t + 6t = 190$ and then rounded the answer to the nearest whole number.

B. Correct. The student demonstrated an ability to solve for a variable in a linear equation.

C. The student ignored the +6 and solved for $8t = 190$.

D. The student added 6 to 190 instead of subtracting.

11

Simone ordered packets of seeds from a store. The packets cost \$2.50 each plus a shipping fee of \$4.00 for the entire order.

Drag the numbers into the table to show how many seed packets Simone can order for each amount in the "Total Cost" column. To place a number in the table, click and hold the number and then drag it to the desired space. To change a number, click and hold it, and then drag it back to the desired space.

Number of Seed Packets Ordered	Total Cost, Including Shipping (\$)
	9.00
	16.50
	21.50
	29.00

1 2 3 4 5

6 7 8 9 10

Standard: 7.A.3.1 Write and solve problems leading to linear equations with one variable in the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are rational numbers.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to write and then solve a problem using an equation with more than one step.

Sample Distractor Rationales:

Correct

Number of Seed Packets Ordered	Total Cost, Including Shipping (\$)
2	9.00
5	16.50
7	21.50
10	29.00

Incorrect

Number of Seed Packets Ordered	Total Cost, Including Shipping (\$)
2	9.00
4	16.50
5	21.50
7	29.00

The student confused the \$2.50 and the \$4.00 and used the equation $y = 4x + 2.50$.

Number of Seed Packets Ordered	Total Cost, Including Shipping (\$)
2	9.00
4	16.50
6	21.50
8	29.00

The student found the correct number of seed packets for \$9.00 and then assumed that the number of seed packets would increase by 2 for each row.

12 **Which situation can be modeled by this inequality?**

$$\boxed{x \geq 12}$$

A Twelve bottles will fill the cardboard box.

B The students had to run at least 12 laps in track practice.

C Use the yardstick to find the length of objects longer than 12 inches.

D The sign says the play area is for children 12 years of age or younger.

Standard: 7.A.3.2 Represent, write, solve, and graph problems leading to linear inequalities with one variable in the form $x + p > q$ and $x + p < q$, where p, and q are nonnegative rational numbers.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to understand what the inequality means and then find a real-life representation of the inequality.

Distractor Rationale:

A. The student confused = and ≥.

B. Correct. The student demonstrated an ability to identify a linear inequality with one variable that represents a real-world situation.

C. The student confused > and ≥.

D. The student confused ≤ and ≥.

13

Match each inequality in the left column with its equivalent inequality in the right column. To make a match, click on an inequality in the left column and then on its equivalent inequality in the right column, and a line will automatically be drawn between them. To remove a connection, hold the pointer over the line until it turns red, and then click it. Each inequality in the left column matches to only one inequality in the right column.

	$x > -6$
$x + 3 < 5$	$x \geq 2$
$x - 1 \geq 9$	$x < 2$
$x + 8 > 2$	$x \leq 8$
$x - 5 \leq 3$	$x < 8$
	$x \geq 10$

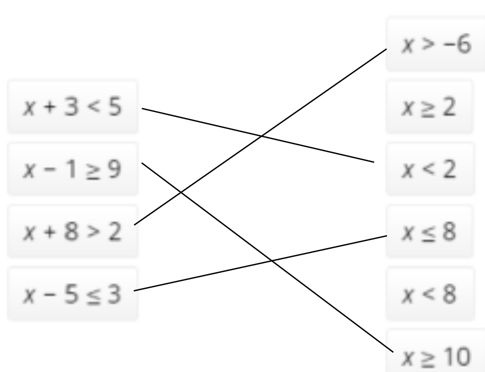
Standard: 7.A.3.2 Represent, write, solve, and graph problems leading to linear inequalities with one variable in the form $x + p > q$ and $x + p < q$, where p, and q are nonnegative rational numbers.

Depth-of-Knowledge: 2

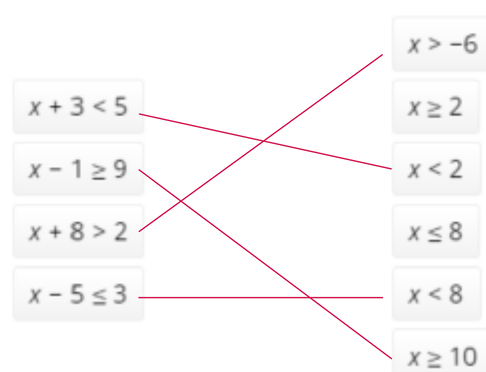
This item is a DOK 2 because it requires the student to decide how to find equivalent inequalities.

Sample Distractor Rationales:

Correct



Incorrect



The student did not understand that $<$ is different from $\leq$.

14 **The expression below will be simplified according to the correct order of operations.**

$$\boxed{9(4 \div 2)^2 + 1 - 6}$$

Which expression results after the first step of simplifying?

A $(36 \div 18)^2 + 1 - 6$

B $9(8 \div 4) + 1 - 6$

C $9(4 \div 2)^2 + 5$

D $9(2)^2 + 1 - 6$

Standard: 7.A.4.1 Use properties of operations (associative, commutative, and distributive) to generate equivalent numerical and algebraic expressions containing rational numbers, grouping symbols and whole number exponents.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, completing the first step when simplifying an expression.

Distractor Rationale:

A. The student started on the right and did 9 × 4 and 9 × 2.

B. The student tried to do the exponent first, but did 4 × 2, not 4^2.

C. The student computed from right to left.

D. Correct. The student demonstrated an ability to apply the order of operations.

15 **A rectangular prism has a square base. The height of the prism is equal to twice the edge length of its base. The base can be completely covered, with no overlap, by 16 squares that each have an edge length of 1 inch.**

What is the total surface area of the rectangular prism?

A 16 square inches

B 32 square inches

C 128 square inches

D 160 square inches

Standard: 7.GM.1.2 Using a variety of tools and strategies, develop the concept that surface area of a rectangular prism can be found by wrapping the figure with same-sized square units without gaps or overlap. Use appropriate measurements (e.g., cm^2).

Depth-of-Knowledge: 3

This item is a DOK 3 because it requires the student to find the surface area of a rectangular prism when given information about the height in relation to the base, rather than giving the actual dimensions.

Distractor Rationale:

A. The student added the dimensions of the prism or found the surface area of only one base.
B. The student found the surface area of the bases only.
C. The student found the volume or the total surface area of the four faces that are not bases.
D. Correct. The student demonstrated an ability to find the surface area of a rectangular prism.

16 **A box is in the shape of a cube. The box has edge lengths of 5 inches. How many unit cubes are needed to fill the box?**

A 25

B 50

C 125

D 150

Standard: 7.GM.1.3 Using a variety of tools and strategies, develop the concept that the volume of rectangular prisms can be found by counting the total number of same-sized unit cubes that fill a shape without gaps or overlaps. Use appropriate measurements (e.g., cm^3).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to know that the length, width, and height are all the same in a cube and then determine volume as the number of cubes required to fill a cube-shaped box.

Distractor Rationale:

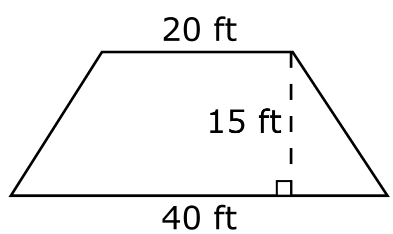
A. The student computed 5 × 5.

B. The student computed 5 × 10.

C. Correct. The student demonstrated an ability to find the number of unit cubes needed to fill a box given the edge length.

D. The student found the surface area of the cube.

17 **The shape and measurements, in feet (ft), of a floor are shown.**



What is the area of the floor, in square feet?

A 190 square feet

B 320 square feet

C 450 square feet

D 900 square feet

Standard: 7.GM.2.1 Develop and use the formula to determine the area of a trapezoid.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding the area of a trapezoid.

Distractor Rationale:

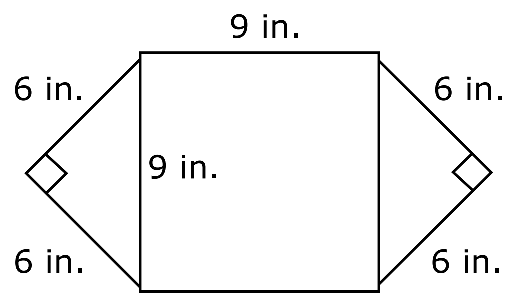
A. Balance distractor

B. The student computed $20 + \frac{40 \times 15}{2}$.

C. Correct. The student demonstrated an ability to determine the area of a trapezoid.

D. The student computed (20 + 40) × 15.

18 **This figure shows the placemat Kenneth made using two square pieces of paper measured in inches (in.). He cut one piece in half.**



What is the area, in square inches, of Kenneth's placemat?

A 42 square inches

B 54 square inches

C 99 square inches

D 117 square inches

Standard: 7.GM.2.2 Find the area and perimeter of composite figures.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to find the area of a composite figure.

Distractor Rationale:

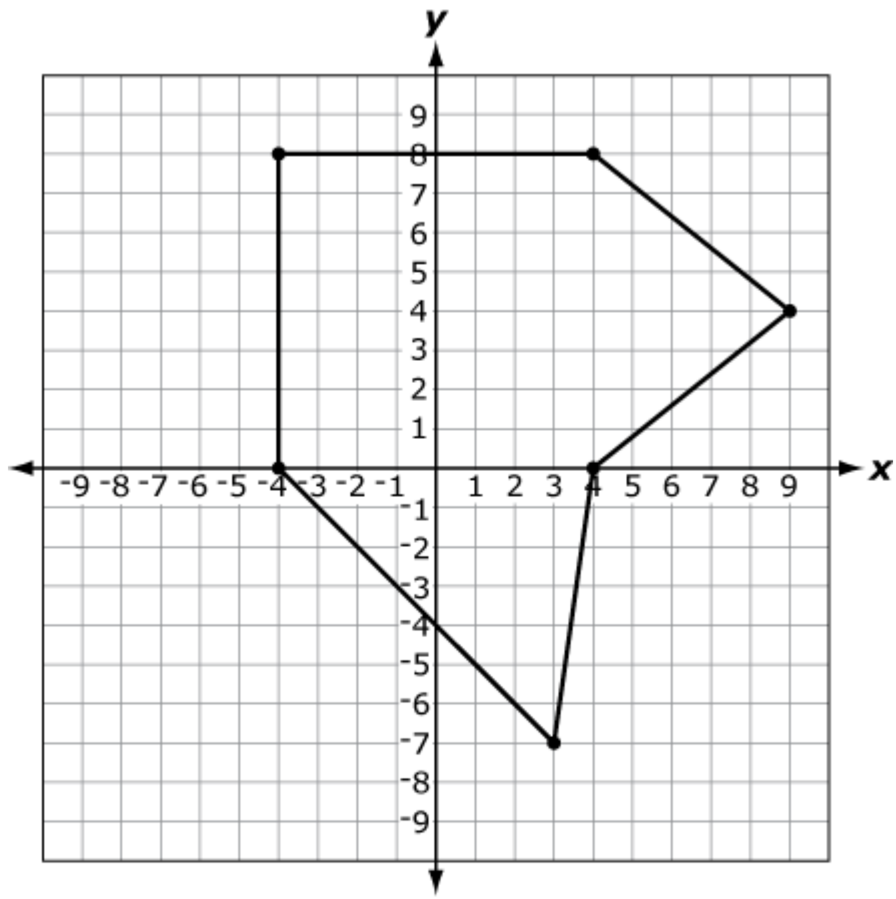
A. The student added all numbers shown on the figure.

B. The student computed 9 × 6.

C. The student focused on the middle square and saw the two 9s.

D. Correct. The student demonstrated an ability to find the area of a composite figure to solve a real-world problem.

19 **The top view of a plot of land is shown on a map, where each unit on the coordinate grid represents 1 meter.**



What is the area of this plot of land in square meters?

A 84 square meters

B 92 square meters

C 112 square meters

D 120 square meters

Standard: 7.GM.2.2 Find the area and perimeter of composite figures.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to decide how to find the area of this composite figure.

Distractor Rationale:

A. The student did not include the triangle on the bottom of the figure.

B. The student did not include the triangle on the right of the figure.

C. Correct. The student demonstrated an ability to find the area of a composite figure to solve a real-world problem.

D. The student thought both triangles were congruent with an area of 28 square meters each.

20 **The circle in the center of a professional basketball court has a diameter of 12 feet. What is the circumference, in feet, of the circle?**

A 6π feet

B 12π feet

C 24π feet

D 144π feet

Standard: 7.GM.3.3 Calculate the circumference and area of circles to solve problems in various contexts, in terms of pi (π) and using approximations for pi (π).

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, finding the circumference of a circle when given the diameter.

Distractor Rationale:

A. The student found πr instead of πd.

B. Correct. The student demonstrated an ability to calculate the circumference of a circle in terms of π.

C. The student found $2\pi d$ instead of πd.

D. The student found πd^2 instead of πd.

21 **Caleb went rollerskating at the local rink. The circular rink measured 70 feet from side to side through the center. What distance, in feet, did Caleb skate in one lap by skating along the edge of the rink?**

A 35π feet

B 70π feet

C 140π feet

D $4{,}900\pi$ feet

Standard: 7.GM.3.3 Calculate the circumference and area of circles to solve problems in various contexts, in terms of pi (π) and using approximations for pi (π).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract the necessary information and then utilize a formula to find the circumference of a circle when given the diameter.

Distractor Rationale:

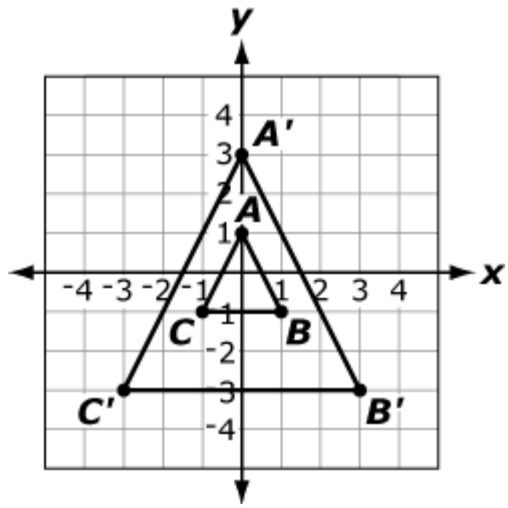
A. The student thought the formula for circumference was πr.

B. Correct. The student demonstrated an ability to calculate the circumference of a circle in terms of π.

C. The student thought the formula for circumference was $2\pi d$.

D. The student thought the formula for circumference was $d^2\pi$.

22 **Triangle *ABC* was dilated with the origin as the center of dilation to create triangle *A*'*B*'*C*'.**



Which scale factor was used to dilate triangle *ABC* to create triangle *A*'*B*'*C*'?

A a scale factor of 2

B a scale factor of 3

C a scale factor of $\frac{1}{2}$

D a scale factor of $\frac{1}{3}$

Standard: 7.GM.4.1 Describe the properties of similarity, compare geometric figures for similarity, and determine scale factors resulting from dilations.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall a simple procedure, identifying the scale factor used in a dilation.

Distractor Rationale:

A. The student saw that it is 2 units from *A* to *A*'.

B. Correct. The student demonstrated an ability to determine scale factors resulting from a dilation.

C. The student saw that it is 2 units from *A* to *A*' and reversed the order of the dilation.

D. The student reversed the order of the dilation.

23 **Jillian made a scale model of her lawn. The actual length of her lawn is 32 feet (ft). The length of her model is 8 inches (in.). What is the ratio of the length of Jillian's lawn to the length of her model?**

A 1 ft : 4 in.

B 4 ft : 1 in.

C 1 ft : 25 in.

D 25 ft : 1 in.

Standard: 7.GM.4.2 Apply proportions, ratios, and scale factors to solve problems involving scale drawings and determine side lengths and areas of similar triangles and rectangles.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to organize the given information in order to determine the relationship between the two lengths.

Distractor Rationale:

A. The student reversed the relationship.

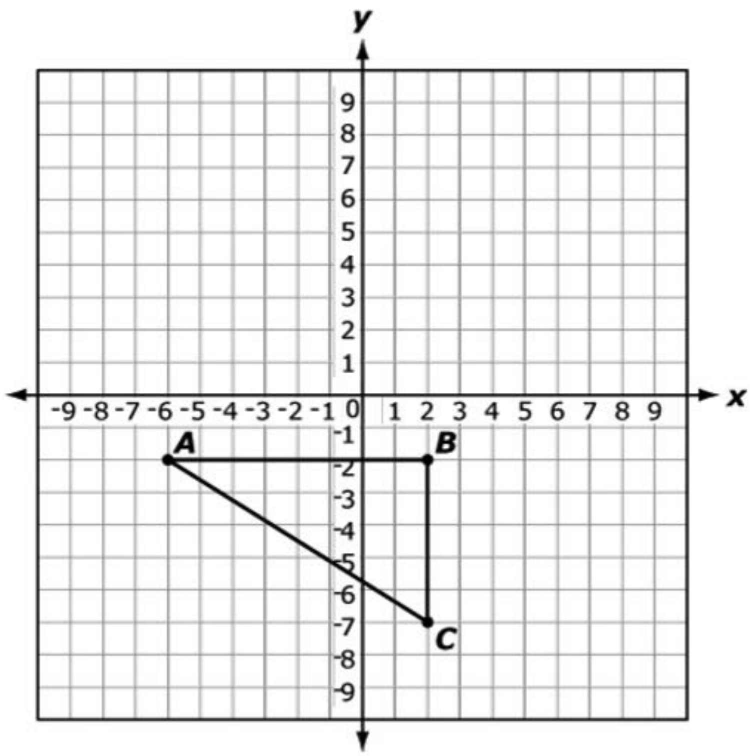
B. Correct. The student demonstrated an ability to apply scale factors to solve a problem involving a scale drawing.

C. The student did $\frac{8}{32} = \frac{1}{4} = 0.25$ and thought this meant 25 inches.

D. The student did $\frac{8}{32} = \frac{1}{4} = 0.25$ and thought this meant 25 feet.

24

Triangle *ABC* is translated 8 units up to create new triangle *A′B′C′*.



Select the ordered pairs that show the coordinates of the vertices of triangle *A′B′C′*.

To select the coordinates for a vertex, click the ordered pair. To deselect the coordinates, click on the ordered pair again.

(2, –2)	(–6, 6)	(–6, –10)	(2, 6)
(2, –10)	(10, –7)	(2, 1)	(10, –2)

Standard: 7.GM.4.3 Graph and describe translations (with directional and algebraic instructions), reflections across the *x*- and *y*-axes, and rotations in 90° increments about the origin of figures on a coordinate plane, and determine the coordinates of the vertices of a figure after a transformation.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to perform a translation and then identify the vertices of the translated figure.

Sample Distractor Rationales:

Correct

(-6, 6) (2, 6) (2, 1)

Incorrect

(2, –2) (10, –2) (10, –7)

The student thought *ABC* and *A'B'C'* were the same triangle.

(–6, –10) (2, –10) (–6, 6)

The student confused 8 units up and 8 units down.

25 **This table shows the number of birds a bird watcher saw each day during a week.**

Bird Watcher Data

Day	Number of Birds
Monday	42
Tuesday	35
Wednesday	31
Thursday	53
Friday	29
Saturday	31
Sunday	52

What is the mean number of birds the bird watcher saw in one day?

A 24

B 31

C 39

D 53

Standard: 7.D.1.1 Design simple experiments, collect data, and calculate measures of center (mean, median, and mode) and spread (range and interquartile range). Use these quantities to draw conclusions about the data collected and make predictions.

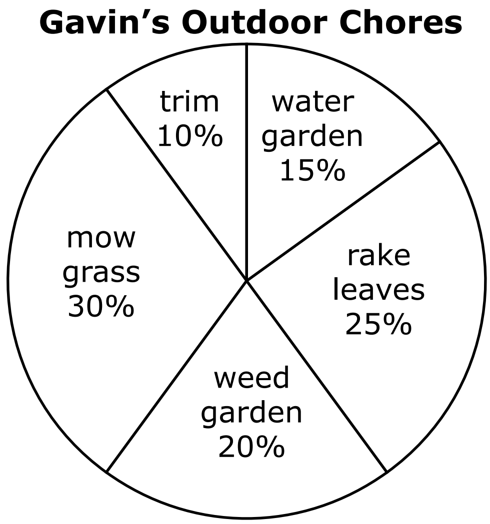
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract information from the table and then utilize a multi-step equation to find the mean of a set of data.

Distractor Rationale:

A. The student confused median and range.
B. The student confused median and mode.
C. Correct. The student demonstrated an ability to calculate the mean for a set of real-world data.
D. The student confused median and the middle number in the table, which is not in order from least to greatest.

26 **Gavin spent 40 hours last month doing outdoor chores. This circle graph shows how much time he spent on each chore.**



How many hours did Gavin spend mowing the grass?

A 8 hours

B 10 hours

C 12 hours

D 30 hours

Standard: 7.D.1.2 Use reasoning with proportions to display and interpret data in circle graphs (pie charts) and histograms.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract information from the graph in order to find 30% of 40 hours.

Distractor Rationale:

A. The student confused 25% and 30%.

B. The student computed 40 − 30.

C. Correct. The student demonstrated an ability to use reasoning with proportions to interpret data in circle graphs.

D. The student thought 30% meant 30 hours.

27

Question 1

At a school carnival game, players toss beanbags onto a table with equal-sized squares of different colors. On the table, there are

- 7 green squares,
- 5 orange squares, and
- 8 blue squares.

These statements describe the probabilities for different outcomes of a single beanbag toss that lands at a random location on the table. **Select the number that best completes each statement.** To select a number, click the menu and then click the desired number. To choose a different number, click the menu and click the new number.

The probability that the outcome is green is -Select an Answer- .

0.28
0.35
0.54
0.7
4
7

The probability that the outcome is orange or green is -Select an Answer- .

$\frac{1}{12}$
$\frac{1}{7}$
$\frac{1}{5}$
$\frac{2}{7}$
$\frac{3}{5}$
$\frac{3}{4}$

The probability that the outcome is <u>not</u> orange is -Select an Answer- .

15%
25%
50%
75%
85%

Standard: 7.D.2.1 Determine the theoretical probability of an event using the ratio between the size of the event and the size of the sample space; represent probabilities as percents, fractions and decimals between 0 and 1.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the probabilities of various outcomes, using different representations (decimal, fraction, and percent).

Sample Distractor Rationales:

Correct

The probability that the outcome is green is 0.35.

The probability that the outcome is orange or green is $\frac{3}{5}$.

The probability that the outcome is not orange is 75%.

Incorrect

The probability that the outcome is green is 0.7.

The probability that the outcome is orange or green is $\frac{1}{12}$.

The probability that the outcome is not orange is 15%.

The student chose 0.7 for green because there were 7 outcomes for green. The student chose 1/12 for orange or green because there were 12 outcomes for orange or green. The student chose 15% for not orange because there were 15 outcomes for not orange.

The probability that the outcome is green is 0.35.

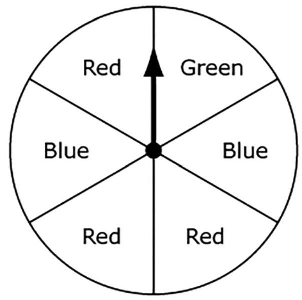
The probability that the outcome is orange or green is $\frac{3}{5}$.

The probability that the outcome is not orange is 25%.

The student gave the probability of orange instead of not orange for the last blank.

28

A spinner is divided into 6 congruent sections and labeled, as shown.



The arrow on the spinner will be spun one time.

Drag and drop an outcome into each box to correctly complete each statement about probability.

To drag an outcome, click and hold the outcome, and then drag it to the desired space. To change an outcome, click and hold it, and then drag it back to the original location. You may use each outcome once or not at all.

blue	green	red	green or red

The probability that the arrow will stop on a section that is labeled [ ] is $\frac{1}{6}$.

The probability that the arrow will stop on a section that is labeled [ ] is $\frac{1}{3}$.

The probability that the arrow will stop on a section that is labeled [ ] is $\frac{2}{3}$.

Standard: 7.D.2.1 Determine the theoretical probability of an event using the ratio between the size of the event and the size of the sample space; represent probabilities as percents, fractions and decimals between 0 and 1.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the probabilities of various outcomes, including combining colors.

Sample Distractor Rationales:

Correct

The probability that the arrow will stop on a section that is **green** is 1/6.
The probability that the arrow will stop on a section that is **blue** is 1/3.
The probability that the arrow will stop on a section that is **green or red** is 2/3.

Incorrect

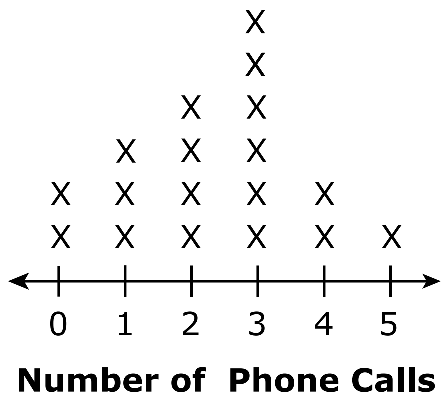
The probability that the arrow will stop on a section that is **green** is 1/6.
The probability that the arrow will stop on a section that is **blue** is 1/3.
The probability that the arrow will stop on a section that is **red** is 2/3.

The student saw three sections for red and focused on the denominator of 2/3.

The probability that the arrow will stop on a section that is green **or red** is 1/6.
The probability that the arrow will stop on a section that is **red** is 1/3.
The probability that the arrow will stop on a section that is **blue** is 2/3.

The student focused on the denominators and chose 1/6 for green or red because that is the greatest denominator and 1/3 for red because that has 3 parts on the spinner. Then the student focused on the numerator and chose 2/3 for blue because there are 2 parts for blue on the spinner.

29 **The line plot shows the number of phone calls made in one day by students in Dorothy's class.**



Based on the information in the line plot, what is the probability a student chosen at random made 2 or 3 phone calls that day?

A $\frac{1}{3}$

B $\frac{1}{10}$

C $\frac{2}{9}$

D $\frac{5}{9}$

Standard: 7.D.2.2 Calculate probability as a fraction of sample space or as a fraction of area. Express probabilities as percents, decimals and fractions.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract information from the line plot in order to find the probability of making 2 or 3 phone calls per day.

Distractor Rationale:

A. The student compared the number of phone calls asked about, 2, to the total number of phone calls shown, 6.
B. Balance distractor
C. Balance distractor
D. Correct. The student demonstrated an ability to express probability as a fraction.

30 **A fair coin is tossed 30 times with the results being either heads or tails. How many times should the result be tails?**

A 15

B 20

C 25

D 30

Standard: 7.D.2.3 Use proportional reasoning to draw conclusions about and predict relative frequencies of outcomes based on theoretical probabilities.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to understand a probability experiment with a fair coin and then determine the expected outcome of tails.

Distractor Rationale:

A. **Correct. The student demonstrated an ability to predict outcomes based on probabilities.**

B. The student made a calculation error or tried to estimate the answer.

C. The student did not understand the probability of fair coin events.

D. The student reported the total number of events.

Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use this information to answer the following questions.

Three expressions are shown.

Expression 1	$\frac{36}{20}$
Expression 2	$5(3x - 1)$
Expression 3	$-\frac{1}{2}(4x - 2)$

31 **Which number is equivalent to Expression 1?**

A $\frac{5}{9}$

B $\frac{9}{5}$

C $\frac{9}{4}$

D $\frac{21}{5}$

Standard: 7.N.1.2 Recognize and generate equivalent representations of rational numbers, including equivalent fractions.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding an equivalent fraction.

Distractor Rationale:

A. The student confused the numerator and denominator.

B. Correct. The student demonstrated an ability to recognize equivalent representations of rational numbers.

C. The student made a multiplication error in the denominator.

D. The student subtracted 15 from both the numerator and denominator.

32 **Which expression is equivalent to the sum of Expression 2 and Expression 3?**

A $7x - 6$

B $7x - 4$

C $13x - 4$

D $13x - 6$

Standard: 7.A.4.1 Use properties of operations (associative, commutative, and distributive) to generate equivalent numerical and algebraic expressions containing rational numbers, grouping symbols and whole number exponents.

Depth-of-Knowledge: 2
This item is a DOK 2 because it requires the student to find the sum of multi-step expressions using the order of the operations and mathematical properties.

Distractor Rationale:

A. The student computed $-\frac{1}{2} \cdot 4x$ as $8x$ instead of $-2x$ and $-\frac{1}{2}x \cdot -2$ as $-1x$ instead of $1x$.

B. The student computed $-\frac{1}{2} \cdot 4x$ as $8x$ instead of $-2x$.

C. Correct. The student demonstrated an ability to use the properties of operations to generate equivalent algebraic expressions.

D. The student computed $-\frac{1}{2} \times -2$ as -1 instead of 1.